



UPCH

# Fundamentos de Biodiseño

## CASO CLÍNICO 9

### ChCC

#### Integrantes:

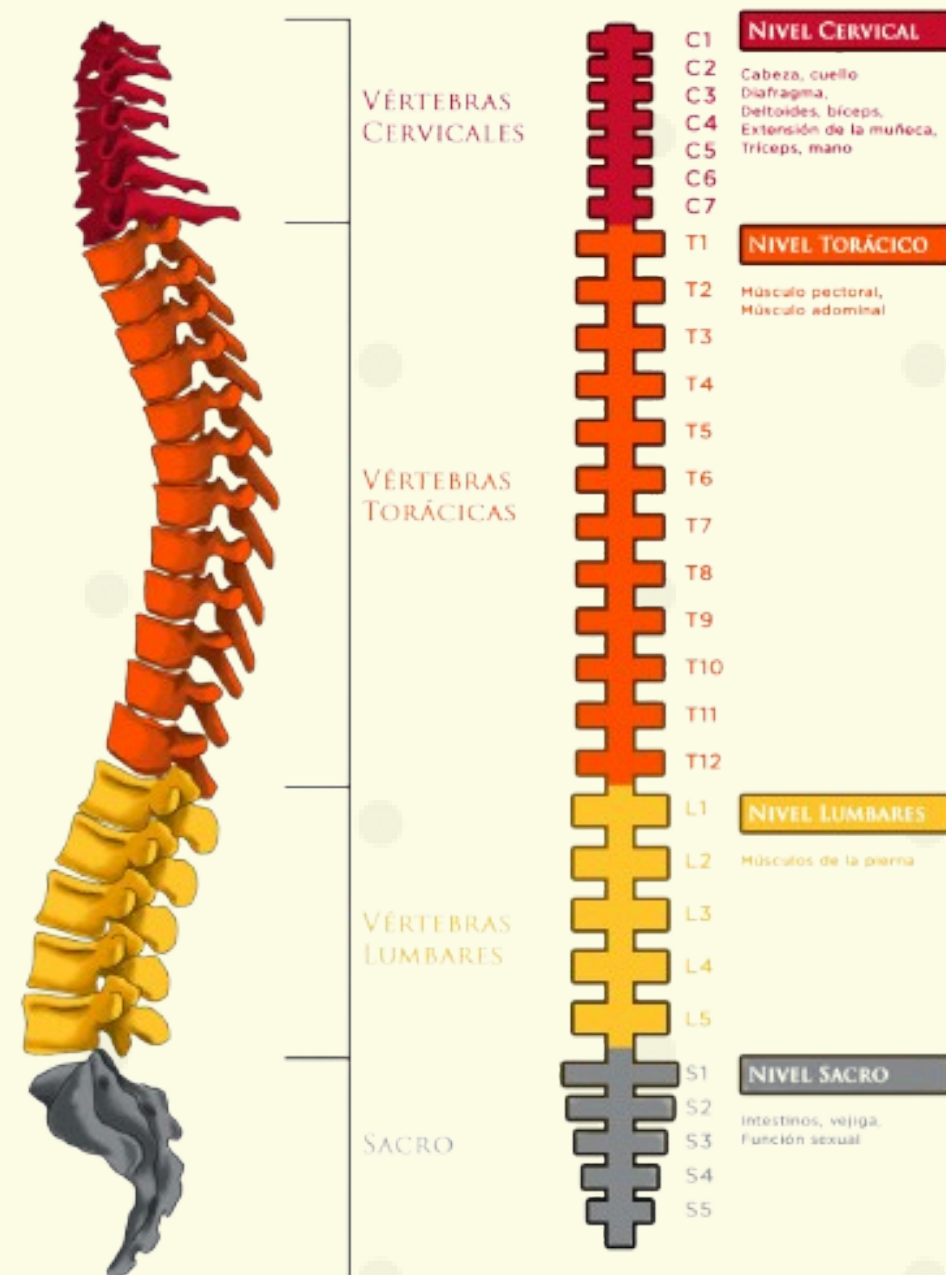
Jaimes, Marcelo  
Salas, Enzo  
Sanchez, Andre  
Sifuentes, Ariana  
Tasaico, Maria  
Zaplana, Bianca



# LESION MEDULAR CERVICA TRAUMATICA

ChCC, 48 años

- 🔫 Proyectoil de arma de fuego
- 🧠 Nivel neurológico: C4
- ⚠️ AIS A — lesión completa



| MOTOR KEY MUSCLES |      | SENSORY KEY SENSORY POINTS |                | SENSORY KEY SENSORY POINTS |                | MOTOR KEY MUSCLES |   |
|-------------------|------|----------------------------|----------------|----------------------------|----------------|-------------------|---|
|                   |      | Light Touch (LT)           | Pin Prick (PP) | Light Touch (LT)           | Pin Prick (PP) |                   |   |
| C2                | 2    | 2                          | 2              | 2                          | 2              | C2                | 2 |
| C3                | 2    | 2                          | 2              | 2                          | 2              | C3                | 2 |
| C4                | 2    | 2                          | 2              | 2                          | 2              | C4                | 2 |
| C5                | 3    | 0                          | 0              | 0                          | 0              | C5                | 3 |
| C6                | 0    | 0                          | 0              | 0                          | 0              | C6                | 0 |
| C7                | 0    | 0                          | 0              | 0                          | 0              | C7                | 0 |
| C8                | 0    | 0                          | 0              | 0                          | 0              | C8                | 0 |
| T1                | 0    | 0                          | 0              | 0                          | 0              | T1                | 0 |
| T2                | 0    | 0                          | 0              | 0                          | 0              | T2                | 0 |
| T3                | 0    | 0                          | 0              | 0                          | 0              | T3                | 0 |
| T4                | 0    | 0                          | 0              | 0                          | 0              | T4                | 0 |
| T5                | 0    | 0                          | 0              | 0                          | 0              | T5                | 0 |
| T6                | 0    | 0                          | 0              | 0                          | 0              | T6                | 0 |
| T7                | 0    | 0                          | 0              | 0                          | 0              | T7                | 0 |
| T8                | 0    | 0                          | 0              | 0                          | 0              | T8                | 0 |
| T9                | 0    | 0                          | 0              | 0                          | 0              | T9                | 0 |
| T10               | 0    | 0                          | 0              | 0                          | 0              | T10               | 0 |
| T11               | 0    | 0                          | 0              | 0                          | 0              | T11               | 0 |
| T12               | 0    | 0                          | 0              | 0                          | 0              | T12               | 0 |
| L1                | 0    | 0                          | 0              | 0                          | 0              | L1                | 0 |
| L2                | 0    | 0                          | 0              | 0                          | 0              | L2                | 0 |
| L3                | 0    | 0                          | 0              | 0                          | 0              | L3                | 0 |
| L4                | 0    | 0                          | 0              | 0                          | 0              | L4                | 0 |
| L5                | 0    | 0                          | 0              | 0                          | 0              | L5                | 0 |
| S1                | 0    | 0                          | 0              | 0                          | 0              | S1                | 0 |
| S2                | 0    | 0                          | 0              | 0                          | 0              | S2                | 0 |
| S3                | 0    | 0                          | 0              | 0                          | 0              | S3                | 0 |
| S4-5              | 0    | 0                          | 0              | 0                          | 0              | S4-5              | 0 |
| TOTALS            | 3    | 6                          | 6              | 8                          | 8              | 3                 |   |
| (50)              | (50) | (50)                       | (50)           | (50)                       | (50)           | (50)              |   |

6

AFECTA

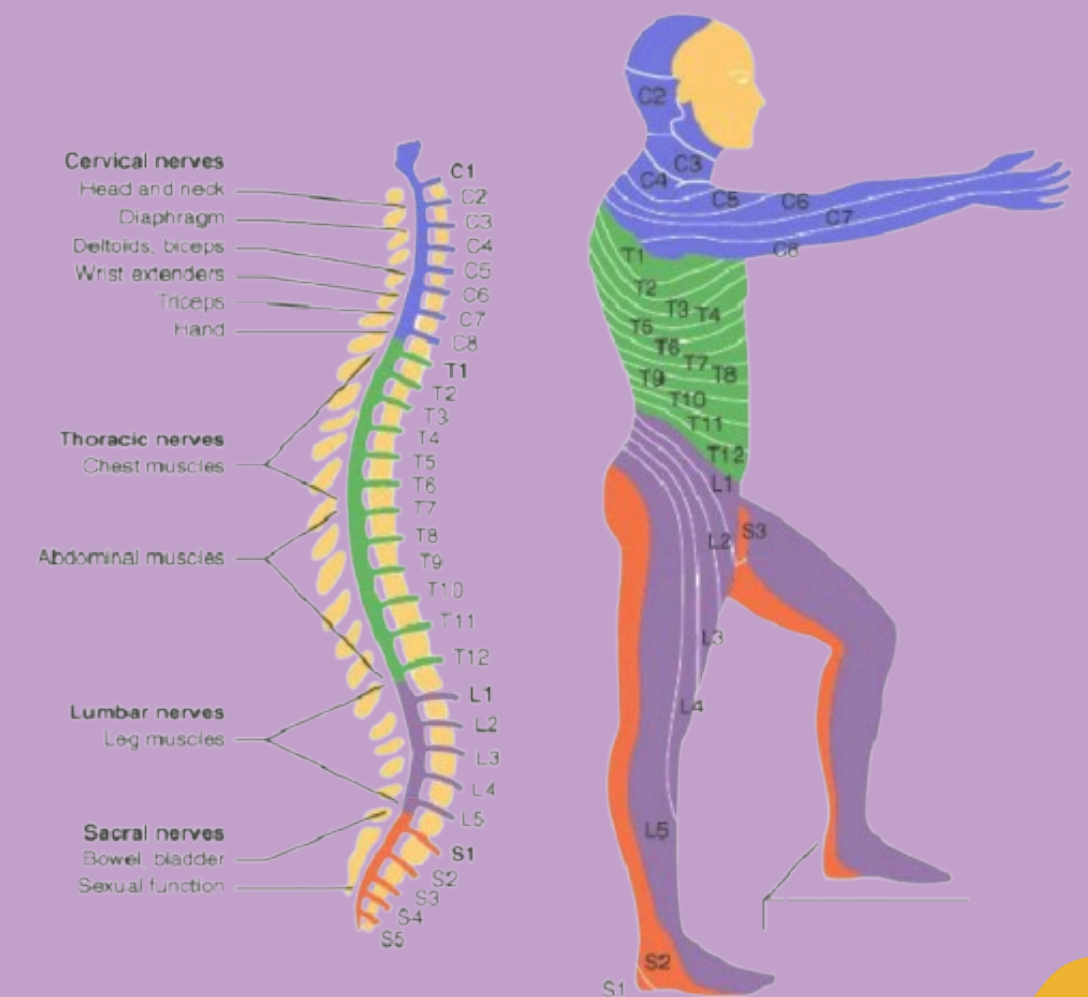
→ SISTEMA NERVIOSO CENTRAL  
→ MÉDULA ESPINAL

ALTERACIÓN COMUNICACIÓN  
CEREBRO-CUERPO

– FUERZA | – SENSIBILIDAD

C5: 3/3 | C6–S1: 0/0

C4 sí | C5–S5 pérdida bilateral



[1] World Health Organization (WHO), "Spinal cord injury," Abr. 2024. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/spinal-cord-injury>.



EQUIPO 14

# Factores de riesgo



➔ Accidentes de tránsito

➔ Caídas

➔ Violencia

➔ Lesiones laborales

➔ Lesiones deportivas

América Central y  
Latinoamérica Tropical  
mostraron proporciones  
relativamente altas de  
lesiones por accidentes de  
tránsito (vehículos  
motorizados) y otros  
accidentes de transporte.

- M. Kim, W. Jeong, S. Jang, J. H. Park, Y. Bae, y S. W. Lee, "Spinal Cord Injury Epidemiology and Causes: A Worldwide Analysis with 2050 Projections," Healthcare, vol. 13, no. 20, Art. no. 2552, Oct. 2025. [Online]. Available: <https://www.mdpi.com/2227-9032/13/20/2552>.
- World Health Organization (WHO), "Spinal cord injury," Abr. 2024. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/spinal-cord-injury>.



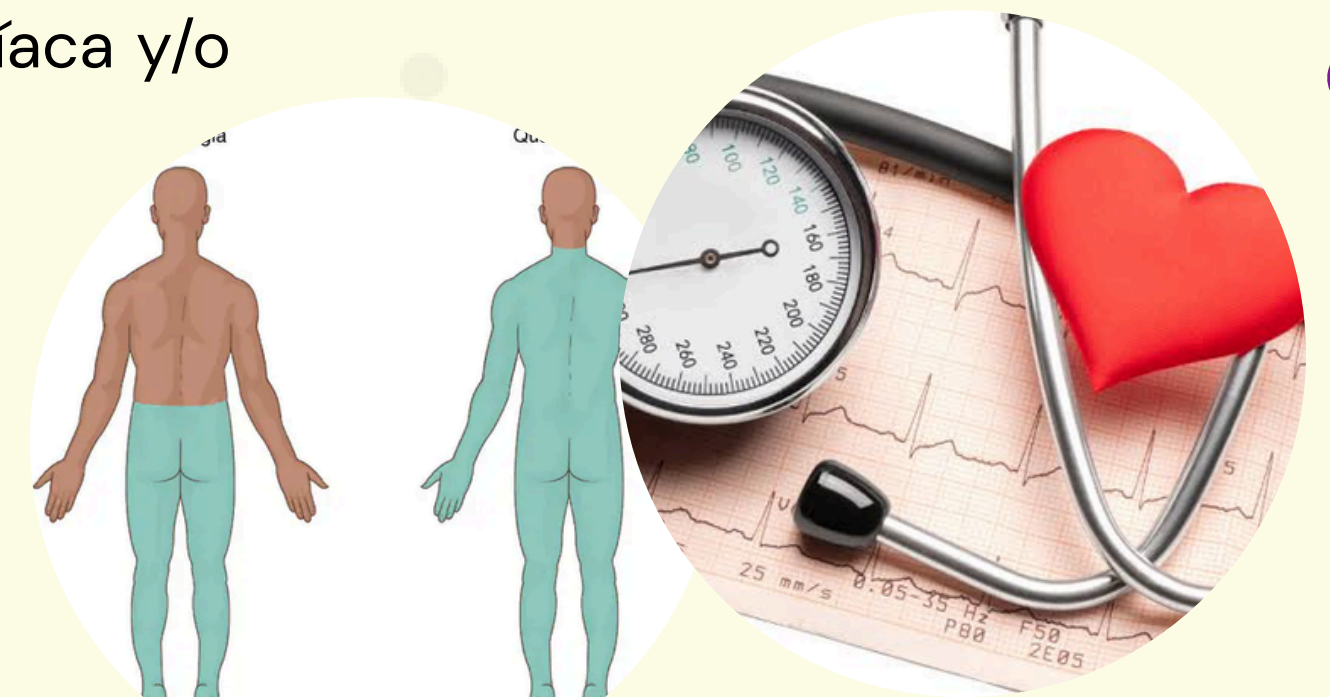
# Manifestaciones clínicas

## SIGNOS Y SÍNTOMAS

- Pérdida parcial o total de las funciones sensoriales y/o motoras (incluidas las funciones de los músculos respiratorios)
- Disfunción intestinal, vesical y sexual
- Desregulación de la presión arterial, la frecuencia cardíaca y/o la temperatura corporal.

## COMPLICACIONES

- Espasticidad
- Dolor (crónico)
- Infecciones del tracto urinario
- Úlceras por presión
- Complicaciones respiratorias
- Disreflexia autonómica
- Trombosis venosa profunda
- Osteoporosis.



# Impacto funcional

Alteración esfinteriana:  
Dependencia para higiene  
y manejo esfinteriano

Perdida de fuerza muscular:  
Movilidad reducida

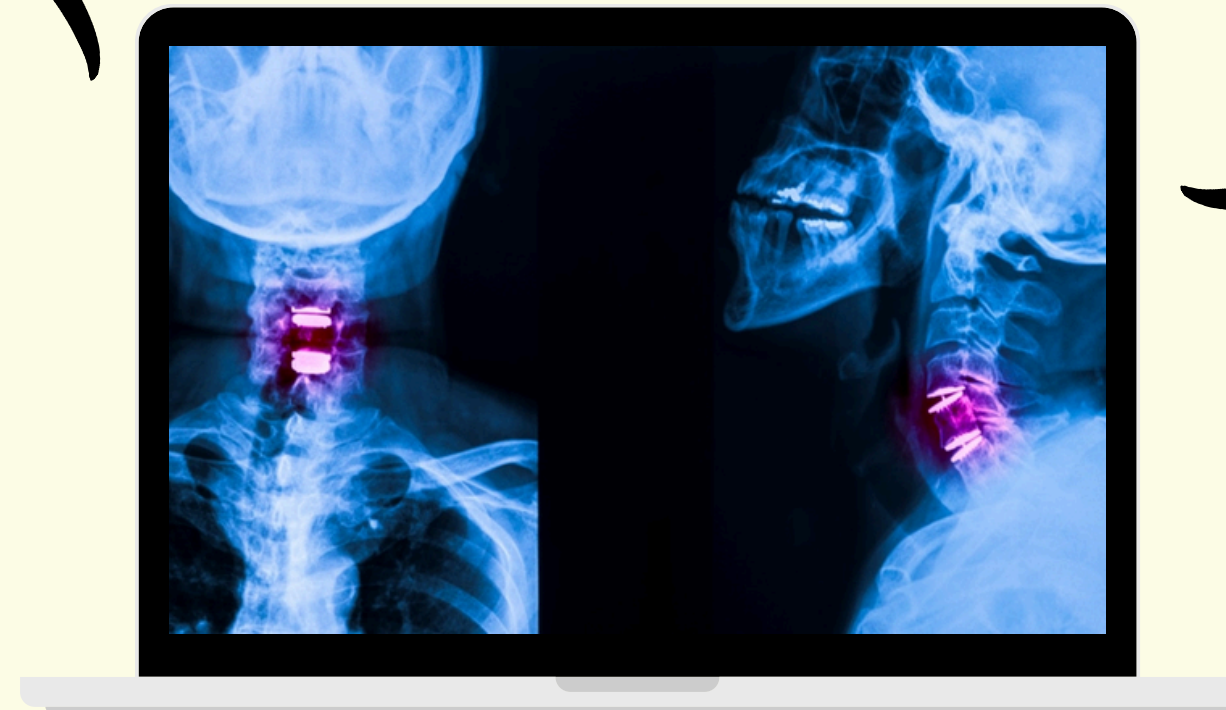


Perdida de sensibilidad:  
Mayor riesgo de lesiones por  
presión

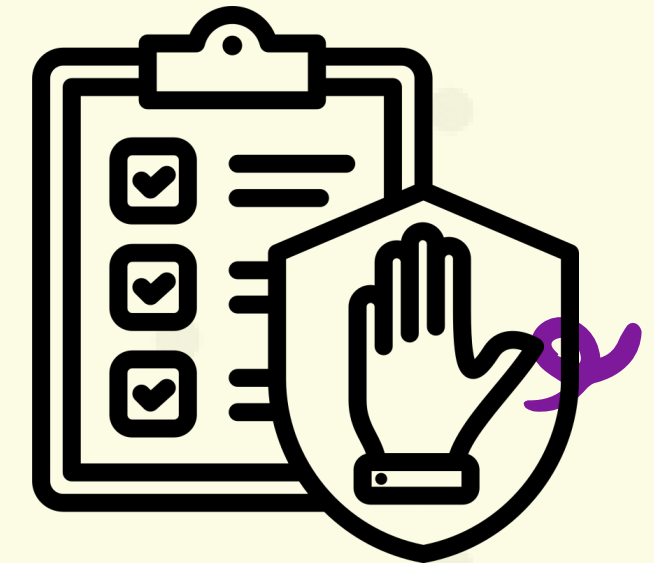
“La paciente presenta dependencia total para las actividades de la vida diaria,  
perdiendo autonomía para desenvolverse de manera independiente.”

# Prevención

90% de LME se deben a causas traumáticas prevenibles



El diseño de medidas preventivas es fundamental



La vigilancia epidemiológica de salud pública es esencial

- World Health Organization (WHO), "Spinal cord injury," Abr. 2024. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/spinal-cord-injury>.
- GBD 2021 Spinal Cord Injury Collaborators, "Global, regional, and national burden of spinal cord injury from 1990 to 2021 and projections for 2050: A systematic analysis for the Global Burden of Disease 2021 study," PubMed, Nov. 2024. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/39603465/>

# Diagnóstico

El diagnóstico inicial exige constatar la ocurrencia de una lesión traumática aguda en los elementos neuronales dentro del canal espinal.

La evaluación clínica debe documentar alteraciones de conciencia o déficits sensitivos, motores y disfunciones autonómicas, ya sean temporales o permanentes.

La confirmación de los casos sospechosos de neurotrauma requiere la revisión de historias clínicas y resultados de estudios por imágenes (como tomografías o resonancias)



- National Notifiable Diseases Surveillance System (NNDSS) / CDC, "Spinal Cord Injury 1990 Case Definition," 1990. [Online]. Available: <https://ndc.services.cdc.gov/case-definitions/spinal-cord-injury-1990/>



# Tratamiento



## Manejo Agudo

Evitar un daño isquémico secundario en un sistema nervioso ya comprometido



## Soporte Farmacológico

Reanimación agresiva con líquidos, monitoreo invasivo



## Rehabilitación

Requieren intervenciones de rehabilitación integrales para recuperar su anatomía.

# Seguimiento



## Seguimiento clínico

Inestabilidad cardiovascular y falla respiratoria, exigiendo un monitoreo crítico constante



## Prevención de complicaciones

Control regular de la función vesical e intestinal para evitar morbilidad secundaria



## Calidad de vida

Resultados funcionales, el nivel de discapacidad y la adaptación psicológica



- World Health Organization Regional Office for the Eastern Mediterranean (WHO EMRO), "Management of Cervical SCI and Neurogenic Shock," Anaesth. Pain Intensive Care, vol. 21, no. 3, pp. 380–382, 2017. [Online]. Available: [https://applications.emro.who.int/imemrf/Anaesth\\_Pain\\_Intensive\\_Care/Anaesth\\_Pain\\_Intensive\\_Care\\_2017\\_21\\_3\\_380\\_382.pdf](https://applications.emro.who.int/imemrf/Anaesth_Pain_Intensive_Care/Anaesth_Pain_Intensive_Care_2017_21_3_380_382.pdf)
- Centers for Disease Control and Prevention (CDC), "Rehabilitation of Persons with Injuries." [Online]. Available: <https://stacks.cdc.gov/view/cdc/188803>





University Presentation

# Necesidades por resolver



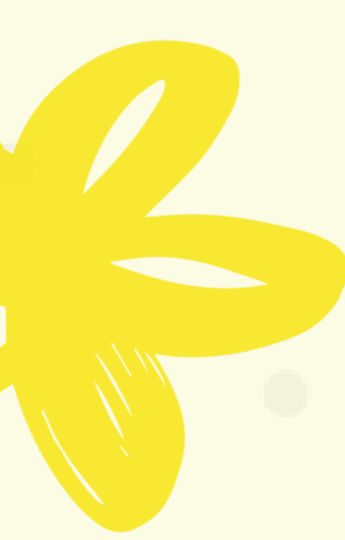
**Control de su cuerpo:**  
Recuperación del control de funciones fisiológicas esenciales, miccionar o excretar, incluso que su cuerpo no pierda masa muscular con el tiempo.



**Restablecer funcionalidad en el día a día:**  
Se plantea el desarrollo de soluciones orientadas a que el paciente recupere autonomía al realizar acciones básicas de la vida diaria.

**Movilidad del paciente:**  
Se propone el diseño de un dispositivo que facilite el desplazamiento del paciente; garantizando su comodidad y desplazamiento autónomo en su entorno.





# Referencias bibliográficas

- [1] World Health Organization (WHO), "Spinal cord injury," Abr. 2024. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/spinal-cord-injury>.
- [2] K. Margetis, J. M. Das, y P. D. Emmady, "Spinal Cord Injuries," *StatPearls Publishing*, Treasure Island (FL), Jun. 2025. [Online]. Available: [https://europepmc.org/article/nbk/nbk560721#\\_article-29326\\_s6](https://europepmc.org/article/nbk/nbk560721#_article-29326_s6).
- [3] M. Kim, W. Jeong, S. Jang, J. H. Park, Y. Bae, y S. W. Lee, "Spinal Cord Injury Epidemiology and Causes: A Worldwide Analysis with 2050 Projections," *Healthcare*, vol. 13, no. 20, Art. no. 2552, Oct. 2025. [Online]. Available: <https://www.mdpi.com/2227-9032/13/20/2552>.
- [4] GBD 2021 Spinal Cord Injury Collaborators, "Global, regional, and national burden of spinal cord injury from 1990 to 2021 and projections for 2050: A systematic analysis for the Global Burden of Disease 2021 study," PubMed, Nov. 2024. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/39603465/>
- [5] National Notifiable Diseases Surveillance System (NNDSS) / CDC, "Spinal Cord Injury 1990 Case Definition," 1990. [Online]. Available: <https://ndc.services.cdc.gov/case-definitions/spinal-cord-injury-1990/>
- [6] World Health Organization Regional Office for the Eastern Mediterranean (WHO EMRO), "Management of Cervical SCI and Neurogenic Shock," *Anaesth. Pain Intensive Care*, vol. 21, no. 3, pp. 380-382, 2017. [Online]. Available: [https://applications.emro.who.int/imemrf/Anaesth\\_Pain\\_Intensive\\_Care/Anaesth\\_Pain\\_Intensive\\_Care\\_2017\\_21\\_3\\_380\\_382.pdf](https://applications.emro.who.int/imemrf/Anaesth_Pain_Intensive_Care/Anaesth_Pain_Intensive_Care_2017_21_3_380_382.pdf)
- [7] Centers for Disease Control and Prevention (CDC), "Rehabilitation of Persons with Injuries." [Online]. Available: <https://stacks.cdc.gov/view/cdc/188803>
- [8] Centers for Disease Control and Prevention (CDC), "Measuring Quality of Life and Function During Early Recovery." [Online]. Available: <https://stacks.cdc.gov/view/cdc/207493>
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